Chapter 7: Architecture Repository

Study Guide, based on Mastering the TOGAF Standard by Eric Jager (2025), Chapter 7 (pp. 127-187)

Overview

Chapter 7 (pp. 127-187) explains the **Architecture Repository**, the Standard's *intangible* central hub where all architecture assets are kept (pp. 127-128). Section 7.1 positions it as a concept, not a tool (pp. 128-129). Section 7.2 walks through its components: Architecture Metamodel, Architecture Landscape (Strategic, Segment, Capability levels), Solutions Landscape, Architecture Requirements Repository, Reference and Standards Libraries, Governance Repository, and Architecture Capability (pp. 129-136). Section 7.3 explains the three **continua** as lenses on the repository (pp. 136-149). Section 7.4 distinguishes catalogs, diagrams, maps, and matrices (pp. 150-154). Section 7.5 covers deliverables, the per-phase artifact list (Table 7-4), and **building blocks** (ABBs and SBBs) (pp. 154-175). Section 7.6 covers views and viewpoints (pp. 176-179). Section 7.7 covers making the repository tangible with a tool (pp. 179-183). Section 7.8 summarizes (pp. 183-187). Leave with this idea: deliverables contain artifacts, artifacts contain building blocks, and building blocks are made of metamodel entities. All of it is stored, classified, and reused through the repository (p. 150). Chapter 7 is part of the GREEN reading tour (p. 127).

> **Reading note:** Repository components, continua, building block types, and views/viewpoints are from the Standard. Jager adds: where ABBs and SBBs are stored, which the Standard does not say (pp. 172, 174-175); the LEGO analogy (p. 164); the binoculars and EHR analogies for views (pp. 177-178); his ZiRA hospital experience (p. 146); a critique of Figure 7-6 (p. 142); and practical tool advice (pp. 179-183). Easy to confuse: the **Architecture Landscape** vs. the **Solutions Landscape**, and a **view** (what you show) vs. a **viewpoint** (the rules for making it).

Detailed Section Breakdown

Chapter Opener (p. 127)

- The repository is "an intangible concept in which all aspects of the architecture are collected and stored". The chapter covers its parts, the continua, building blocks, views and viewpoints, and making it tangible with tools (p. 127).

7.1 The Central Hub (pp. 128-129)

- **Not a tool.** Organizations often treat their tools (from spreadsheets to full applications) as the repository, but in TOGAF the repository is an intangible concept (p. 128).
- **What it should contain.** Only reusable building blocks, an organization-specific metamodel, and architecture governance information. "A well-designed repository does not contain diagrams of specific solutions" (p. 128).
- **Backbone of the practice.** It brings frameworks, templates, and reference material together, for finished and in-progress work, and feeds the perspectives of the three continua (p. 128, Figure 7-1 on p. 129).

7.2 Key Components of the Architecture Repository (pp. 129-136)

Component	What it holds	Page
Architecture Metamodel	How artifacts are organized and how information types relate. Based on the Enterprise Metamodel (Ch. 5)	pp. 129-130
Architecture Landscape	Views of the current, planned, and aspirational architecture at three levels (below), plus reusable ABBs	pp. 130-131
Solutions Landscape	Models and documents of individual Solution Architectures, and reusable SBBs. Not the content/data the solutions produce	pp. 131-132
Architecture Requirements Repository	All requirements created or updated during the ADM, at Strategic, Segment, and Capability levels	pp. 132-134
Reference Library	Best practices, guidelines, templates, patterns, reference architectures, and the Viewpoint Library	p. 134
Standards Library	Standards and policies the architecture must conform to (legal/regulatory, industry, organizational)	pp. 134-135
Governance Repository	Decisions, rules, principles, approvals, waivers, compliance assessments. Provides accountability and traceability	p. 135
Architecture Capability	Skills, processes, structures, and roles that support EA	pp. 135-136

- **Three levels of granularity (Figure 7-2):**

Level	Purpose	Used by
Strategic Architecture	Long-term summary view of the whole organization, for direction setting	Executives (p. 130)
Segment Architecture (also "Domain Architecture")	More detailed operating models for areas of the organization	Program or portfolio level (p. 130; Ch. 10.3.2)
Capability Architecture	Detail on supporting a unit of capability: current, target, and increments. Groups work packages into portfolios and programs	Project/work package level (p. 130)

- These three levels matter for agile architecture (Ch. 18) (p. 131).
- **Requirements by level.** Strategic (executive, long-term direction), Segment (portfolio/project, tactical breakdown of strategic requirements), Capability (needs of individual initiatives or work packages) (pp. 132-133).
- **External inputs.** Both libraries are fed by external reference models (e.g., BIZBOK) and external standards (e.g., the TOGAF Standard itself) (p. 135).
- **Architecture Board.** Gives visibility to the Governance Repository and Architecture Capability, serves as an escalation mechanism, and directs the Architecture Capability (p. 136).

7.3 The Three Continua (pp. 136-149)

- **What a continuum is.** A lens or conceptual framework for viewing and classifying the repository's assets, scaling detail, complexity, and governance to need. Left to right gives an increasingly detailed view (pp. 136-137).

Continuum	Role	Page
Enterprise Continuum	Umbrella over the other two. Bridges conceptual architectures and practical solutions. Enables reuse	p. 138
Architecture Continuum	Classifies architecture assets from generic to specific. Applies to the Architecture Landscape, Reference Library, and Standards Library	pp. 138, 141

Continuum	Role	Page
Solutions Continuum	Parallel to the Architecture Continuum. Shows how architectures become deployable solutions. Applies to the Reference Library and Solutions Landscape	pp. 138, 147

7.3.1 The Enterprise Continuum (pp. 139-141)

- Shows evolution "from generic to specific, from abstract to concrete, and from logical to physical". It is the outermost continuum, classifying contextual assets (policies, standards, strategic initiatives, structures, organization-level capabilities) and also solutions (pp. 139-140).
- Its scale runs from foundation (most general) to organization-specific (Figure 7-5, p. 140).

7.3.2 The Architecture Continuum (pp. 141-146)

Type	Description	Healthcare example	Page
Foundation Architectures	Most generic. Universal services and technologies (network protocols, OS, security frameworks, SOA)	n/a	pp. 142-143
Common Systems Architectures	Generic across industries. Complete for one problem domain (security, networking, etc.)	Patient management, scheduling, billing systems	p. 143
Industry Architectures	Reference models and standards for one industry	EHR, telemedicine, HIE, HL7, ICD-11	pp. 143-144
Organization-Specific Architectures	Most tailored: one organization's goals, processes, culture	A hospital's customized EHR with a patient portal and local pharmacy integration	p. 144

- **Four steps (Figure 7-7):** start with Foundation, adopt Common Systems, tailor with Industry, customize with Organization-Specific (p. 145).

Jager's commentary (author's view): Figure 7-6 is hard to read, because Foundation and Common Systems parts are not visibly reused in the later models (p. 142). In Dutch hospitals he reused **ZiRA** (the Dutch Hospital Reference Architecture), an Industry Architecture, which saved enormous time (p. 146).

7.3.3 The Solutions Continuum (pp. 147-149)

Type	Description	Page
Foundation Solutions	Highly generic concepts, tools, products, services (including professional and support services)	p. 148
Common Systems Solutions	Implementations of a Common Systems Architecture, e.g., an enterprise management or security product	pp. 148-149
Industry Solutions	Reusable industry-specific packages, e.g., a physical database schema or point-of-service device	p. 149
Organization-Specific Solutions	Implementations of the Organization-Specific Architecture, with the most unique content	p. 149

- **Two directions (Figure 7-9).** Left to right delivers solution value. Right to left addresses business needs (p. 148).
- **Solutions inventory.** A populated Solutions Continuum is a solutions inventory. Connecting the two continua aims to build organization-specific solutions on generic ones (pp. 147, 149).

7.4 Catalogs, Diagrams, Maps, and Matrices (pp. 150-154)

- **Hierarchy (Figure 7-10):** a deliverable contains one or more artifacts, an artifact contains one or more building blocks, and a building block contains one or more entities (p. 150).

Form	Definition	Example	Page
Catalog	List of building blocks of one entity type, for governance or reference. Entities are *not* linked	Stakeholder Catalog	p. 151
Map	Textual representation of architecture content	Business Capability Map, Organization Map	pp. 151-153
Diagram	Graphical representation of the same kind of content	Process Flow Diagram	pp. 151-153
Matrix	Grid of relationships between two or more entities ("cross-mapping")	Entity A vs. Entity B	p. 153

- **Less is more.** Better to create more views than to overload one artifact (p. 154).

7.5 Architecture Deliverables (pp. 154-175)

- Deliverables support conversations about architecture issues. Example: an Architecture Definition Document containing an Organization Map, a Process Flow Diagram, and an Application Landscape (pp. 154-155, Figure 7-12).

7.5.1 Architecture Artifacts (pp. 155-163)

- **Not all are required.** "Creating all the artifacts listed in the TOGAF Standard is by no means a necessity." Use those that add value (pp. 155, 157).
- Artifacts are derived from the tailored metamodel and created in the Preliminary Phase, Requirements Management, and Phases A-E (p. 156, Figure 7-13).

Phase	Recommended artifacts (Table 7-4)
Preliminary	Principles Catalog (p. 157)
A: Architecture Vision	Stakeholder Catalog; Value Chain Diagram; Solution Concept Diagram; Business Model Diagram; Business Capability Map; Value Stream Map (p. 157)
B: Business	Catalogs: Organization/Actor; Driver/Goal/Objective; Role; Business Service/Function; Location; Process/Event/Control/Product; Contract/Measure; Business Capabilities; Value Stream; Value Stream Stages; Business Glossary. Matrices: Business Interaction; Actor/Role; Value Stream/Capability; Strategy/Capability; Capability/Organization. Diagrams and maps: Business Footprint; Business Service/Information; Functional Decomposition; Product Lifecycle; Goal/Objective/Business Service; Business Use-Case; Organization Decomposition; Process Flow; Business Event; Business Capability Map; Value Stream Map; Organization Map; Information Map (pp. 158-160)
C: Data	Data Entity/Data Component Catalog; Data Entity/Business Function Matrix; Application/Data Matrix; Conceptual Data, Data Dissemination, Data Security, Data Migration, and Data Lifecycle Diagrams (p. 160)
C: Application	Application Portfolio and Interface Catalogs; Application/Organization, Role/Application, Application/Function, and Application Interaction Matrices; Application Communication, Application and User Location, Application Use-Case, Enterprise Manageability, Process/Application Realization, Software Engineering, Application Migration, and Software Distribution Diagrams (pp. 161-162)
D: Technology	Technology Standards and Technology Portfolio Catalogs; Application/Technology Matrix; Environments and Locations, Platform Decomposition, Processing, Networked Computing/Hardware, and Network and Communications Diagrams (pp. 162-163)
E: Opportunities and Solutions	Project Context Diagram; Benefits Diagram (p. 163)

7.5.2 Building Blocks (pp. 163-175)

- **Definition.** "A potentially reusable component that can be combined with other building blocks to deliver architectures and solutions". They are packages of functionality corresponding to metamodel entities, with clear boundaries, and should not be tightly bound to an implementation (pp. 165-166).
- **LEGO analogy (Figure 7-14).** Generic bricks combine into bigger sets. Three SBBs stack into an ABB (pp. 164-165).
- **Level of detail grows.** A block may begin as a name and become detailed later. Sometimes less detail is more effective, e.g., one Business Capability Map instead of a 100-page document (pp. 165-167).

7.5.2.1 Building blocks in architecture design (pp. 167-168). An architecture is "a set of building blocks depicted in an architectural model and a specification of how those building blocks are connected". Only include blocks that address the problem. Blocks interact in complex ways, and one can support many or partially support one. Blocks start general and become precise, and must follow standards and Enterprise Principles (pp. 167-168).

7.5.2.2 Specification process (pp. 168-169, Figure 7-16).

Phase	What happens to building blocks
A	High-level model of candidate blocks, e.g., a Business Capability Map
B, C, D	Specify the entities that flesh out that map. Reuse existing blocks from the repository, create missing ones via catalogs/matrices/diagrams, and remove obsolete ones
E	Work packages (initiatives) are defined to create the missing blocks
Overall	ABBs are identified for goals, then iteratively refined into SBBs that are bought or built

(pp. 169)

7.5.2.3 Types (p. 170). ABBs = the *what and why* (technology-agnostic). SBBs = the *how* (specific technologies and products). SBBs normally *implement* ABBs, but when the ABB is a capability and the SBB a data, application, or technology entity, the SBB can also *automate* it (Figure 7-17).

7.5.2.4 Architecture Building Blocks (pp. 171-173). Created in the initial ADM iteration. They capture requirements and capabilities, specify fundamental functionality, attributes, interoperability, and relationships, and map to business or organizational entities. Example: a Human Resources Management capability. Stored (per Jager) in the Architecture Landscape, Reference Library, and Standards Library (Figure 7-18).

7.5.2.5 Solution Building Blocks (pp. 173-175). Detailed, product- or vendor-aware, procured or developed. They map to IT topology and operational policies, with attributes such as security, manageability, scalability, performance, and configurability. Example: the Candidate Evaluation Service. Stored (per Jager) in the Solutions Landscape (Figure 7-19).

	ABB	SBB
Answers	What and why	How
Technology	Agnostic	Product/vendor-aware
Created	Early (initial iteration, A-D)	Later, refined from ABBs
Stored (Jager)	Architecture Landscape, Reference Library, Standards Library	Solutions Landscape

7.6 Views and Viewpoints (pp. 176-179)

- **View.** A representation of a system from the perspective of specific stakeholders, addressing their concerns, e.g., a security view (p. 176).
- **Viewpoint.** The perspective and rules (a template) for creating a view, e.g., a performance viewpoint (p. 176).
- **Relationship (7.6.1).** A viewpoint is a blueprint, and many views can come from one viewpoint. "A view is what you actually present... a viewpoint is the set of rules" for creating it. Both live in the Reference Library (p. 177).
- **Binoculars analogy (Figure 7-20).** Two people on two peaks look at one village. People = stakeholders, binoculars = viewpoints, village = view (pp. 177-178).
- **EHR example (Figure 7-21).** Physician and patient see the same EHR system differently, so multiple viewpoints are needed (p. 178).

7.7 Using the Architecture Repository (pp. 179-183)

- **Make it tangible with a tool.** Office tools soon fail: no unified modelling language, and every artifact must be tracked manually. A good tool must relate entities so changes propagate (pp. 179-180).

7.7.1 Deciding on a Tool (pp. 180-183)

- Budget drives the choice. Gartner's yearly report helps. A tool is "an absolute must" to make the repository tangible (p. 180).
- **Table 7-5 considerations (p. 181):** modelling language support (full ArchiMate, BPMN, user-defined properties, coloured views); metamodel customization; flexibility and stability (change one concept into another while keeping relationships, handle large models); import/export (share with non-users, publish large PDFs as posters); support (help desk, user community).
- Publishing artifacts to a website is "an easy step to higher maturity" (p. 182).
- **Minimum components to represent (Figure 7-22):** Architecture and Solutions Landscapes, Reference and Standards Libraries, Architecture Requirements Repository (p. 182).
- "A well-managed repository is characterized by ruthlessly minimizing information collected and maintained" (p. 183).
- **Reality check.** Nine out of ten tools don't separate the Architecture and Solutions Landscapes. Use an intuitive folder structure to do so (p. 183).

7.8 Summary (pp. 183-187)

- Three continua (Enterprise as umbrella, Architecture, Solutions), each with four types that become more detailed from left to right (pp. 183-184).
- Catalogs, diagrams/maps, and matrices visualize building blocks. Artifacts consist of blocks, and deliverables consist of artifacts (p. 184).
- Building blocks come in ABB and SBB flavours, are defined mainly in A-D, and are stored in the repository. Views and viewpoints communicate them. A tool is essential (pp. 185-187).

Key Terms & Definitions

Term	Definition	Page
Architecture Repository	Intangible central hub storing reusable building blocks, the organization-specific metamodel, and governance information	p. 128

Term	Definition	Page
Architecture Landscape	Current, planned, and aspirational architecture at Strategic, Segment, and Capability levels	p. 130
Solutions Landscape	Solution Architectures and reusable SBBs	p. 131
Governance Repository	Record of decisions, approvals, waivers, and compliance assessments	p. 135
Enterprise Continuum	Umbrella lens showing evolution from generic to specific, abstract to concrete, logical to physical	pp. 138-140
Architecture Continuum	Foundation, Common Systems, Industry, Organization-Specific Architectures	pp. 141-144
Solutions Continuum	Foundation, Common Systems, Industry, Organization-Specific Solutions	pp. 147-149
Catalog / Map / Diagram / Matrix	List of one entity type / textual representation / graphical representation / relationship grid (cross-mapping)	pp. 151-153
Architecture deliverable	Product of architecture work made of one or more artifacts	pp. 150, 154
Building block	Potentially reusable component combined with others to deliver architectures and solutions	p. 165
ABB / SBB	Technology-agnostic what-and-why block / product-aware how block that implements (or automates) an ABB	pp. 170-174
View / Viewpoint	Representation for stakeholders' concerns / rules or template for creating views	p. 176

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Self-Check Questions

Q: Is the Architecture Repository a tool? What should it contain?

A: No, it is an intangible concept. It should contain reusable building blocks, an organization-specific metamodel, and governance information, but not diagrams of specific solutions (p. 128).

Q: Name the three levels of the Architecture Landscape and who uses each.

A: Strategic (executive direction, whole organization), Segment/Domain (program or portfolio), and Capability (units of capability, grouping work packages and projects) (p. 130).

Q: Name the four types in the Architecture Continuum, from generic to specific.

A: Foundation, Common Systems, Industry, Organization-Specific Architectures (p. 141).

Q: What do left-to-right and right-to-left movements on the Solutions Continuum represent?

A: Left to right: delivering solution value by building on more generic solutions. Right to left: addressing business needs (p. 148).

Q: What is the difference between a catalog and a matrix?

A: A catalog lists building blocks of one entity type without linking them. A matrix cross-maps relationships between two or more entities (pp. 151, 153).

Q: How are deliverables, artifacts, building blocks, and entities related?

A: A deliverable has one or more artifacts, each artifact has one or more building blocks, and each building block has one or more entities (p. 150).

Q: Distinguish ABBs from SBBs.

A: ABBs describe the technology-agnostic what and why (capabilities, requirements). SBBs describe the how: specific products and technologies that implement, or for capabilities also automate, ABBs (pp. 170-174).

Q: In which ADM phases are building blocks mainly specified, and what happens in Phase E?

A: Mainly in A, B, C, and D. In Phase E, work packages are defined to create the missing building blocks (p. 169).

Q: Explain views and viewpoints using the village analogy.

A: The people on the peaks are stakeholders, their binoculars are viewpoints (the rules and perspective), and the village is the view (what is seen) (pp. 177-178).

Q: Must every artifact in Table 7-4 be produced?

A: No. The Standard only shows possibilities. Produce the artifacts that add value and ignore the rest (pp. 155, 157).

Q: Which repository components should an architecture tool minimally support?

A: The Architecture and Solutions Landscapes, the Reference and Standards Libraries, and the Architecture Requirements Repository (p. 182).

Q: What did Jager use ZiRA for, and which continuum type is it?

A: As a reusable Dutch hospital reference architecture (business function and care process models) when doing hospital EA. It is an Industry Architecture (p. 146).